



QP CODE: 2242203950



2242203950

Reg No :

Name :

MCA DEGREE EXAMINATIONS, JULY 2022

Third Semester

Master of Computer Application

Core - MCACT301 - MACHINE LEARNING TECHNIQUES

2020 Admission Onwards

D3124AAD

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Explain about the issues related to machine learning.
2. Explain about the application of Machine learning.
3. Differentiate most specific hypothesis and most general hypothesis with an example.
4. What is Prior Probability and Posterior Probability ?
5. Write the weakness of the decision tree.
6. What is feature extraction?
7. Explain about Linear Discriminant Analysis?
8. What are the disadvantages of K- means clustering?
9. Define Dendrogram with an example.
10. Define the concept of voting.
11. What are the various area where Artificial Neural Network can be used?
12. Explain about the application of Neural Network.

(10×3=30 marks)

Part B

*Answer **all** questions*

*Each question carries **9** marks*

13. a) Write short note on i) Classification ii) Regression iii) Clustering





OR

b) Discuss about the terms i) Machine Learning ii) Deep Learning

14. a) i) Discuss about Bayes decision theory

ii) The following data set contains factors that determine whether tennis is played or not. Using Naive Bayes classifier, find the play prediction for the day <Sunny, Cool, High, Strong>

DAY	OUTLOOK	TEMP	HUMIDITY	WIND	PLAY
Day 1	Sunny	Hot	High	Weak	NO
Day 2	Sunny	Hot	High	Strong	NO
Day 3	Overcast	Hot	High	Weak	YES
Day 4	Rain	Mild	High	Weak	YES
Day 5	Rain	Cool	Normal	Weak	YES
Day 6	Rain	Cool	Normal	Strong	NO
Day 7	Overcast	Cool	Normal	Strong	YES
Day 8	Sunny	Mild	High	Weak	NO
Day 9	Sunny	Cool	Normal	Weak	YES
Day 10	Rain	Mild	Normal	Weak	YES
Day 11	Sunny	Mild	Normal	Strong	YES
Day 12	Overcast	Mild	High	Strong	YES
Day 13	Overcast	Hot	Normal	Weak	YES
Day 14	Rain	Mild	High	Strong	NO

OR

b) i) Discuss about Simple Linear Regression

ii) Use the following data construct a linear regression model and evaluate the regression coefficients.

X	55	60	65	70	80
Y	52	54	56	58	62

15. a) Express your idea about Feature Extraction and Feature Selection methods used in Machine Learning.

OR

b) Explain about Linear discriminant analysis.

16. a) Describe AGNES clustering algorithm.

OR





b) Describe DBSCAN algorithm for clusters.

17. a) Explain the Structure of an Artificial Neural networks.

OR

b) What is Backpropagation feedforward neural network? Explain

(5×9=45 marks)

